Sampling for the rCARI Validation Study - Burkina Faso

- rCARI Study Team - WFP, ROME -

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1 Background

The rCARI validation study is a global research project commissioned by WFP and funded by the USAID Bureau of Humanitarian Affairs (BHA) to investigate and compare remote and face-to-face (F2F) modules of the consolidated approach for reporting food insecurity (CARI). The study runs in six pilot countries.

The overall goal of the study is to understand to what extent food insecurity reported by remote CARI (rCARI) is comparable with the standard CARI collected through face-to-face interviews. Both remote and F2F data collection modules & methodologies for collecting food insecurity will be examined in the study.

This document outlines the study methodology, specifically the sampling process adjusted for country specific context of Burkina Faso. The study has been designed and implemented as a survey experiment with stratified sample selected across 2 regions (ADMIN1) in each pilot country.

2 Overall study sampling design

The following considerations underpin the study areas and final sample selection.

- Phone ownership & accessibility within the pilot country
- Distribution of food insecurity
- Areas of accessibility taking adjusting for difference in livelihoods

It is important to emphasize that the purpose of data collection for this study is not to produce population-level estimates but rather to compare the two methodologies within a statistically comparable population. While it is desirable to collect data from a variety of livelihood zones in both urban and rural communities to assess the impact of these differences with respect to modality, national coverage and representativeness is not the primary objective of the current study.

Figure 1 below presents the generic study sampling procedure. As mention earlier, selection and inclusion of study area followed a 3-point criterium: i.e., 1) phone coverage, 2) accessibility, and 3) food security situation. Phone coverage is an important consideration allowing for equal probability of reaching an equal number of households for both remote and F2F interviews, while field accessibility ensures that enumerators can successfully access and interview households in face-to-face. The food security consideration would ensure that areas that have different AFI (acute food insecurity) situations are adequately represented in the study sample.

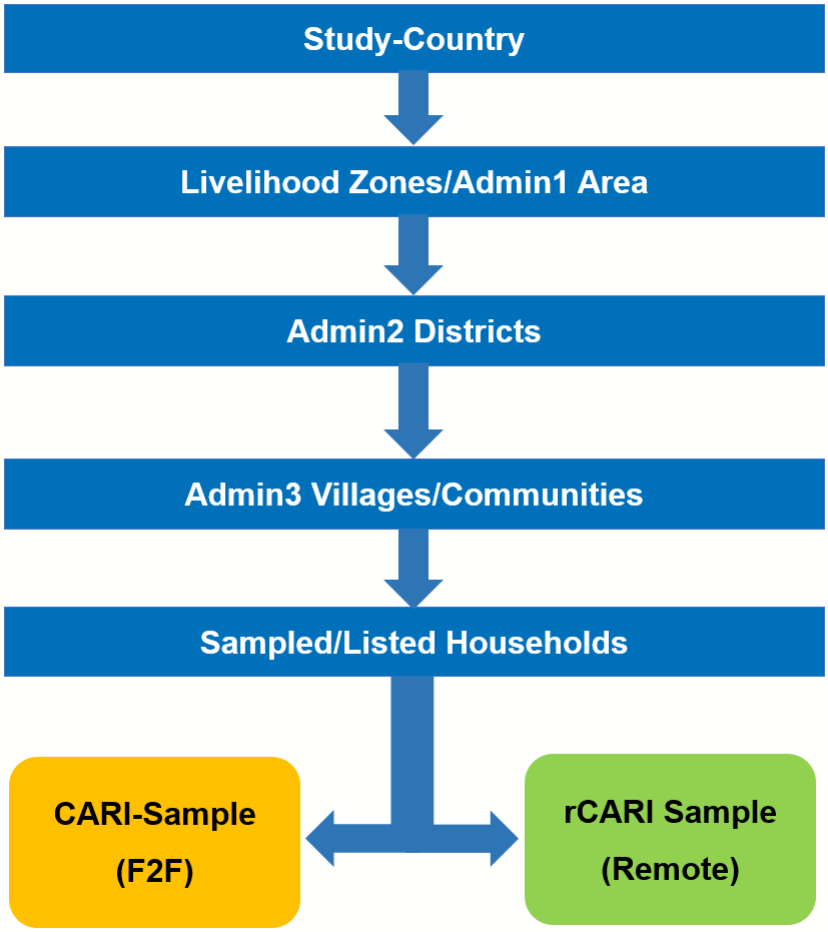

Figure 1: Sampling and Randomization Procedure

The main steps followed in identifying and selecting the admin areas at the various stages are listed and explained below.

Step 1 - Selection of ADMIN1 (Livelihood zones)

The first stage involves the identification and selection of Admin1 or livelihood areas that meets the inclusion criteria. A minimum of 2 Admin1 Areas will be purposively identified and selected (on the recommendation WFP country offices) based on phone coverage, food security situation and accessibility. If more than 2 Admin1 Areas meet the inclusion criteria, a simple random approach will be used to select 2 Areas.

Step 2 - Selection of ADMIN2

In the second stage, a minimum of 4 ADMIN2 areas are randomly selected (2 per ADMIN1 Area based on the outcome of Step 1) from the ADMIN1 areas that have been identified and selected.

Step 3 - Selection of ADMIN5 - (Lowest Admin Areas i.e., villages)

After identifying and selecting the Admin2 areas, all villages (ADMIN5 areas) in the selected ADMIN2 areas that met the inclusion criteria set out in Step 1 are listed as primary sampling units (PSUs). A full census of PSUs will be conducted in each selected ADMIN2 area to ensure equal chance of selection once the inclusion criteria is met. A list of 20 PSUs will be randomly selected (distributed equally across the selected ADMIN1s from Step 1). The study sample (households) will be selected from these 20 PSUs.

Step 4 - Sampling of households (HHs)

The final stage of the sampling procedure involves enlisting eligible households into the study. The sampling frame for the study includes all households in each of 20/25 selected PSUs. Sampling households for the study follows a simple random sampling approach (in the absence of any recent household census, there will be a fresh census conducted by the CO team with support from HQ). The aim is to collect information from households that is as accurate and representative as possible in each study area. Thus, the idea is to give equal chance of selection to all households in each PSU as much as practicable. The study will then randomly select an equal number of households (minimum of 150 households) per PSU taking into consideration previous attrition and experience of the CO. Trained and conscientious enumerators will be used for the household selection process.

Informed consent will be taken from all households at the listing phase to ensure that all households understand and agree to participate in the study. Based on the number of PSUs per country, around 3000HHs will be sampled for the study. Only sampled households will be used for the randomization in the next stage.

Step 5 - Randomization of household (HHs)

Consented final sampled households in Step 4 will be randomly assigned into one of two survey modalities/groups; F2F (CARI group) and Remote (rCARI group). The design is such that the sample will be divided equally between the two groups using a standardized randomization process.

All interviews (both F2F & remote) will be conducted simultaneously over the same time frame to ensure that households are reporting food security situation over the same window and also affected by similar geographical situations.

2.1 Burkina Faso pilot

Following the study framework illustrated in Figure 1, in conjunction with the Burkina Faso Country Office (CO), the first three steps have been completed. Two prefectures (ADMIN1s), namely Plateau Central, Centro-Nord and Nord, have been purposively selected for the study based on accessibility and phone coverage.

Since 2021, Burkina Faso has faced significant security challenges from violent extremist groups opposing the central government. A risk assessment conducted with the CO in 2023 indicated that Burkina Faso remains exposed to several security threats from these groups, which may impact the ability to conduct onsite face-to-face (F2F) interviews and build rapport over the phone for successful interviews. Despite these challenges, some level of normalcy in terms of security has been maintained in the Sahel, Plateau Central, Centro-Nord, Nord, and Est regions. Since 2021, the WFP, in collaboration with partners such as UNHCR, local leaders, and the government, has implemented extensive protocols to mitigate security risks in these regions, improving access and operations for local partners. This has significantly enhanced access to Centro-Nord and Nord, where the WFP has a strong collaborative presence.

Regarding phone accessibility, the two selected regions have a mobile phone penetration rate of between 85-110 percent, which supports the feasibility of conducting remote data collection for the study.

2.1.1 ADMIN1 AREAS (3 Prefectures)

1. North Prefecture, and
2. Centro-Nord Prefecture
3. Plateau Central Prefecture

These Admin1 areas have been carefully selected (purposively) to ensure that they meet the inclusion criteria of phone access, accessibility given the prevailing security conditions in BF and food security situation. Inclusion in particular was based on the latest Cadre Harmonise (CH) food security analysis conducted in March 2024. Furthermore, these are have been selected because of the possibility of reaching many households with varied and diverse livelihood activities as much as possible.

2.1.2 ADMIN3 AREAS (4 Communes)

Four Admin3s in total were then selected from the ADMIN1s i.e., 1 in Centro-North, 1 in Plateau Central and 2 in Nord Prefectures. The selection of communes has been done with special consideration to allow for a mix of rural and urban households this will include livelihood activities prevalent in rural and urban Burkina Faso such as formal employment, self-employment and agricultural activities. The ADMIN3 (Communes) selected are listed below.

1. Meguet
2. Korsimoro
3. La-Todin and,
4. Oula

2.1.3 ADMIN5 AREAS (37-Villages)

Based on a census of all ADMIN5 areas in the Plateau Central, Nord and Centro-Nord Prefectures (Selected ADMIN1s) conducted by the Burkina Faso CO, 37 villages (Admin5 areas) have been selected to form the main PSUs from which participating households would be drawn.

Selection of the ADMIN5 areas followed a simple random sampling procedure drawing villages randomly from a census of villages in each of the 4 Admin3 areas. The distribution of the selected 37 villages are as follows: 10 villages from Plateau Central Prefecture 8 villages from the Centro-Nord Prefecture and, 19 villages from Nord Prefecture.

Table 1 shows the list of ADMIN5 areas selected with approximate size (number of households). The number of households per village are based on data from 2021 and may be slightly different on the ground at the time of the study. Based on the current acute food insecurity numbers (using the March 2024 CH analysis), Table 1 presents the final list of villages that have been selected for the study in the three ADMIN1 Areas.

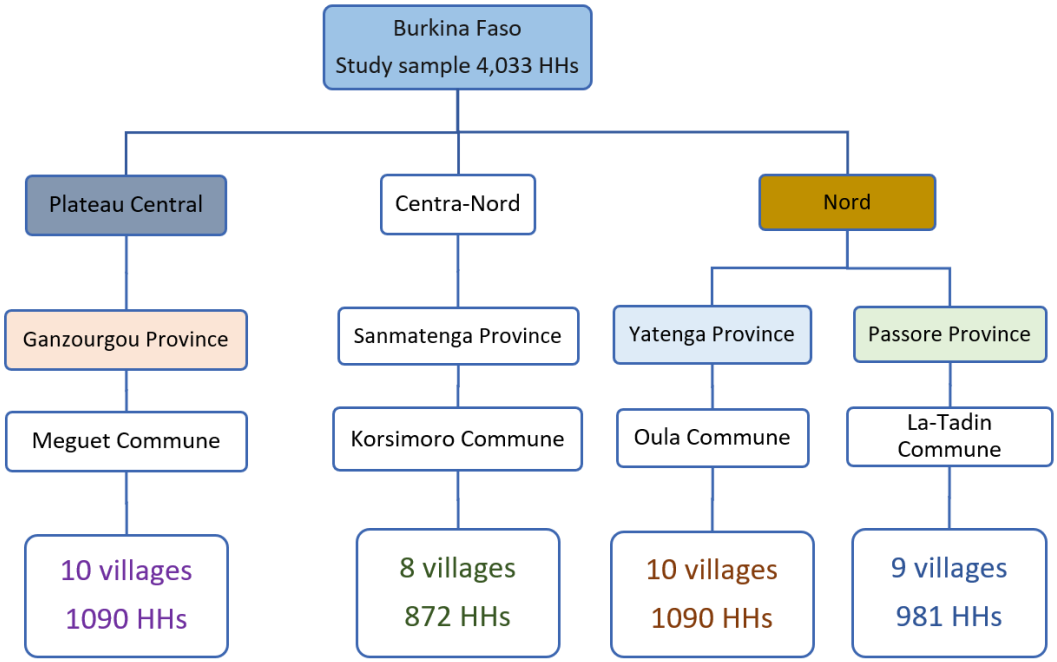

Figure 2: Systematic sampling process - ADMIN1 to Respondents

Table 1: List of Selected ADMIN5 Areas with Total Population for BF Pilot

REGION	PROVINCE	COMMUNE	Village	Population	CH Class
PLATEAU CENTRAL	GANZOURGOU	MEGUET	Kougdoughin	240	IPC1
			Boulwando	460	IPC1
			Baghin	248	IPC1
			Nahoubé	270	IPC1
			Zemalga	623	IPC1
			Koulweogo	396	IPC1
			Tamasgo	397	IPC1
			Tibin	425	IPC1
			Vagma	255	IPC1
			Kabouda	360	IPC1
CENTRE NORD	SANMATENGA	KORSIMORO	Ilartenga	188	IPC3
			Nigui	193	IPC3
			Ouitenga-Poecin	210	IPC3
			Bendogo	249	IPC3
			Nimpoui	264	IPC3
			Tamboko	293	IPC3
			Noungou	408	IPC3
			Ipalla	469	IPC3
NORD	PASSORE	LA-TODIN	Minissia	355	IPC2
			Bissiga	329	IPC2
			Gogho	325	IPC2
			Baribsi	208	IPC2
			Kingria	592	IPC2
			Sissamba	254	IPC2
			Kollo	396	IPC2
			Loungo	852	IPC2
			Kalamtogo	259	IPC2
NORD	YATENGA	OULA	Bougoure	596	IPC3
			Bourbo	235	IPC3
			Pelle	379	IPC3
			Omsom	412	IPC3
			Oula	313	IPC3
			Lougouri	647	IPC3
			Zanna Mossi	236	IPC3
			Sonh	229	IPC3
			Tilli	220	IPC3
			Kayegue Mossi	248	IPC3

2.1.4 Selection of respondents (4033 Households)

The sampling frame in Burkina Faso includes all households in the 37 selected villages (PSUs). Households will be enrolled in the study through a simple random sampling procedure, drawing respondents from a comprehensive census of households in each of the 37 selected villages.

Carefully selected and trained enumerators will conduct the household listing in each PSU. Due to the lack of recent census data for the selected areas (according to the CO), a full household census will be conducted in each village during a pre-survey phase. During this phase, households will be contacted, consent for participation will be obtained, and baseline and contact information will be collected. A standardized listing questionnaire will be administered to gather basic household and respondent characteristics such as the household head's age, gender, education, household composition, ownership of basic assets, mobile phone ownership, housing, and WASH. These characteristics will be used later for randomization purposes.

The final sampling frame will include households that have consented, have access to mobile phones, and have completed the listing questionnaire. Sampling of actual study participants will follow a simple random process based on the final cleaned census list.

To ensure a uniform distribution of the sample across the study areas, an equal number of households (109 households) will be sampled from each village, resulting in a final sample size of 4,033 households.

2.1.5 Randomizing the study sample

Households enlisted that do not have access to mobile phones or phone coverage will be excluded during the randomization process. Similarly, households with incomplete baseline and contact information will be dropped from the final sample. Randomization will be conducted separately for each ADMIN5 sample to ensure comparability in the livelihood responses reported across modalities in each ADMIN5 area.

The randomization process to assign the final sample is presented in Figure 3.

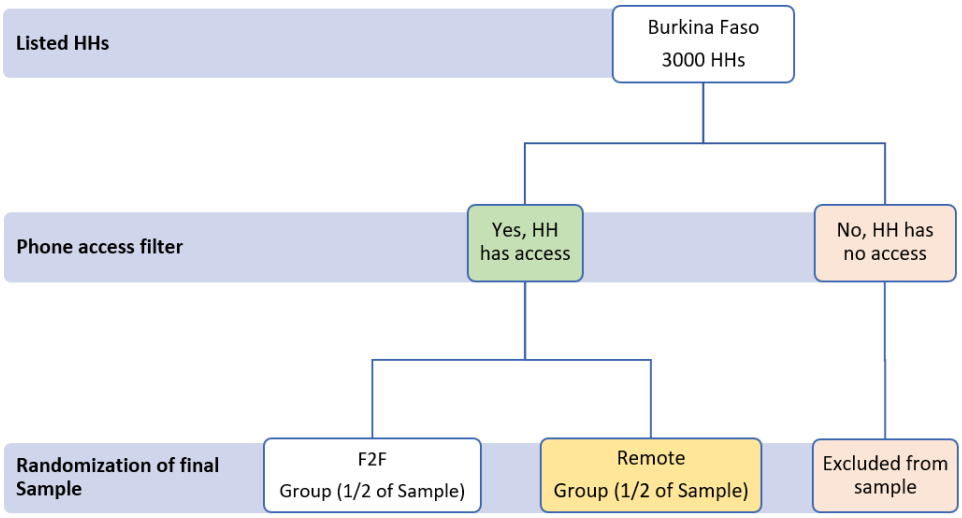

Figure 3: Step by step randomization of the survey sample

2.2 Training & assignment of enumerators

To be qualified to be engaged as an enumerator for both the listing and data collection, all potential enumerators should have;

1. At least a university degree,
2. First hand knowledge of the study areas (preferably should speak the local language/dialect), and
3. Prior data collection experience (at least must have been involved in two separate food security surveys) in BF.

To ensure consistency in data collection, a 5-day in person training will be organized for both F2F and remote enumerators. Training will be conducted jointly by the BF CO (VAM team), RBD and rCARI team from HQ. Training will be largely conducted in French, the predominant language of the study area. Local dialects will be used in cases where required to increase comprehension. A day's hands-on field pilot will be conducted at the end of the training session to test both the practical knowledge of enumerators and interview instruments for both cohorts (F2F and remote). A comprehensive field guide translated into French will be prepared at the end of the enumerator training. This will be provided to all enumerators to serve as quick reference in the field when they are in doubt. In addition, field supervisors with adequate understanding of food security and livelihoods of the study area will be engaged to work closely with all enumerators in the field to increase data quality.

2.3 Survey protocols to enhance response rate

Before, during, and after the survey, several protocols will be established to ensure the successful collection of high-quality data. These protocols are designed to address the challenges associated with low response rate and to promote transparency, trust, and participation of households.

Firstly, collaboration with local authorities and influential individuals in each village (ADMIN5s) will initiated and facilitated by WFP Country Office (CO) and survey contractors. This engagement is aimed to sensitize potential respondents to the survey, fostering transparency and increasing trust. By involving key stakeholders at the local level, the survey team will be able to garner support and cooperation from the local community, laying the foundation for successful data collection.

Secondly, a stand-alone phase household listing/census phase will allow enumerators to inform and sensitize households of the upcoming interviews. This proactive approach will ensure that households are aware of the survey and give them the opportunity to choose a convenient time to be interviewed. By providing advance notice and allowing households to schedule interviews at their convenience, the risk of ambushing respondents or calling them at inconvenient times was minimized, enhancing the likelihood of participation and cooperation. Overall, these pre-survey protocols are essential for establishing rapport with the community, building trust among

respondents, and ensuring that data collection proceed smoothly and effectively.

In addition to the pre-survey protocols aimed at community/village engagement and scheduling, a comprehensive calling protocol has also been developed in collaboration with the hunger monitoring team of the World Food Programme (WFP). Leveraging their expertise in remote data collection, this protocol was designed to ensure consistency, accuracy, and professionalism during phone interviews. The calling protocol outlined guidelines and procedures for enumerators to follow when conducting interviews remotely. These guidelines covered various aspects of the interview process, including:

1. **Introduction:** Enumerators will be instructed on how to introduce themselves and the survey to respondents in a clear and professional manner, fostering trust and cooperation from the outset.

2. **Consent:** Procedures for obtaining informed consent from respondents will be outlined, emphasizing the importance of respecting respondent privacy and autonomy throughout the interview process.

3. **Questioning Techniques:** Enumerators will be trained on effective questioning techniques to elicit accurate and comprehensive responses from respondents while maintaining a respectful and non-biased approach.

4. **Data Recording:** Guidelines for accurately recording respondent answers and maintaining data integrity will be provided to ensure consistency and reliability in the collected data.

5. **Ethical Considerations:** Enumerators were reminded of the ethical considerations involved in conducting phone interviews, such as confidentiality and data security, and instructed to adhere to ethical standards throughout the survey process.

Training sessions for enumerators included a detailed review of the calling protocol, with practical exercises and simulations to reinforce understanding and proficiency. By equipping enumerators with clear guidelines and best practices for conducting phone interviews, the calling protocol aimed to enhance the quality and reliability of the collected data, ultimately contributing to the success of the survey.

2.4 Tentative study timelines & milestones

Table 2: Tentative timelines for the field work

L'heure / Activity	April - May		July – August 2024						
			W1	W2	W3	W4	W5	W6	W7
1. *Questionnaire adaptation to BF context*	Completed								
2. *Census of households in ADMIN5*									
3. *Training of Enumerators*									
4. *Testing of Questionnaire*									
5. *Data Collection*									
6. *Final Dataset & Report*									

Note W=Week

2.5 Possible limitations of the design & suggested mitigation strategies

Although a lot of though has been given to the methodological design of the study, it must be mentioned that there remain some potential unavoidable sources of biases in this current experimental design setup. The potential sources and how the current design would remedy or reduce them are discussed below.

1. Lack of national representativeness of the survey. One of the key limitations of the methodology is its lack of nationwide coverage which is likely to affect the generalization of the results. However, given the main objective of the study, the sample design has been adapted in such a way to ensure that the study sample will representative and comparable as possible. This will ensure that the two study cohorts are drawn from the same sample and comparable.

2. Geographical variation in the distribution of respondents. With the traditional phone survey likely to have lower response rates than F2F, there is a likelihood that completed surveys may be skewed in favour of F2F group which can affect the final analysis. To address this limitation and ensure that the final interview sample is large enough to allow for statistical analysis, a number of strategies will be adopted. First, the study will only be rolled out in areas that allow for both F2F and remote surveys. Second, a comprehensive listing (census) phase has been incorporated to ensures that we are able to achieve a uniform pre-survey distribution of both F2F and remote cohorts. Third, increase sensitization of the study regions through the listing phase. A standard questionnaire will be used for the census that will pre-inform all households in selected PSUs about the study including time frames that they may be contacted for the respective interviews. Forth, increase the number of follow-up in the remote cohort. Specifically, in addition to the stated time by respondents to contacted, 3 additional time windows, i.e., mornings, afternoons and evenings will be used to make calls. This will ensure that more households are reached within the survey period.

3. Confounding through enumerator effects (gender, training and motivation). Given that both survey cohorts are to be collected simultaneously, the same enumerator cannot be used to collect both cohorts which can inadvertently influence the results. To address this potential bias, randomization will be adopted when assigning enumerators to the PSUs and modalities (a minimum of two enumerator will be assigned to each PSU). Also, to ensure equivalence of comprehension of the questionnaire across all enumerators, F2F and remote enumerators will be trained together by the same facilitators. Also, knowledge test will be conducted at the end of the enumerator training with the lowest scoring enumerators being or retrained or dropped all together. Finally, daily frequency speed checks will be put in place by the data management team to be based in Rome access the performance of each enumerator. Those with questionable data inputs will either be asked to re-do the interviews or where it is extreme are dismissed at together. Finally, a conscious effort will be made to increase the number of female enumerators conducting the remote interviews (evidence show that female enumerators tend to achieve higher response rates compared to males in phone survey).

4. High attrition particularly in the remote cohort. High attrition rates particularly in the remote sample can affect the statistical power and the ability to do further statistical analysis using sub groups within the completed surveys. To minimise this limitation, the team has increased initial proposed sample to 4000 households in Burkina Faso. This is expected to create adequate allowance to take care of sample attrition. It must however be mentioned that this will come cost implications for the study.